

1 • Gauss + superposition

- $\oint \vec{E} \cdot d\vec{a} = Q_{enc}/\epsilon_0$. Sphere / coaxial cylinder / pillbox so E is constant and $\parallel d\vec{a}$.
- Ball: $\vec{E} = \frac{\rho}{3\epsilon_0} \vec{r}$ in, $\frac{kQ}{r^2} \hat{r}$ out. $V_{in} = \frac{kQ}{2R^3}(3R^2 - r^2)$, $V(0) = \frac{3}{2}V(R)$.
- Cylinder: $\frac{\rho}{2\epsilon_0} s \hat{s}$ in, $\frac{\lambda}{2\pi\epsilon_0 s} \hat{s}$ out ($\lambda = \rho\pi a^2$). Line: $V = -\frac{\lambda}{2\pi\epsilon_0} \ln \frac{s}{s_0}$; only ΔV .
- Shell: $E = 0$ in. Sheet: $\frac{\sigma}{2\epsilon_0}$. Metal surface: $\frac{\sigma}{\epsilon_0}$.
- **Hole** = $+\rho$ **plus** $-\rho$. Each piece's field from **its own** center/axis; add vectors.
- Ball cavity offset $\vec{d}$: $\vec{E} = \frac{\rho}{3\epsilon_0} \vec{d}$ uniform. Spherical cavity in a cylinder: $\frac{\rho}{\epsilon_0}(\frac{x}{6}, \frac{y}{6}, -\frac{z}{3})$, not uniform.
- Ring axis $E_z = \frac{kqz}{(z^2 + R^2)^{3/2}}$. Disk $V = \frac{\sigma}{2\epsilon_0}(\sqrt{z^2 + R^2} - z)$.

2 • Delta functions, $E \rightarrow \rho$

- $\int f \delta(x-a) dx = f(a)$, $\delta(kx) = \frac{\delta(x)}{|k|}$, $\delta^3(\vec{r})$ has units 1/vol. Shell: $\rho = \sigma \delta(r-R)$.
- $\nabla \cdot \frac{\hat{r}}{r^2} = 4\pi\delta^3(\vec{r})$, $\nabla^2 \frac{1}{r} = -4\pi\delta^3(\vec{r})$. Flux of $\hat{r}/r^2$ through any sphere is 4π , but the divergence is 0 for $r \neq 0$, so it is all at the origin.
- $\rho = \frac{\epsilon_0}{r^2} \frac{d}{dr}(r^2 E_r)$ ($r \neq 0$). $Q_{enc}(r) = 4\pi\epsilon_0 r^2 E_r$.
- **Origin**: $\lim_{r \rightarrow 0} 4\pi\epsilon_0 r^2 E_r = q \neq 0 \Rightarrow +q \delta^3(\vec{r})$. ($\rightarrow 0$: none. $\rightarrow \infty$: infinite charge, unphysical.)
- Jump in E_r at $R \Rightarrow \sigma = \epsilon_0(E_{out} - E_{in})$. $Q_{tot} = \lim_{\infty} 4\pi\epsilon_0 r^2 E_r = \text{point} + \text{volume} + \text{surface}$.
- Poisson $\nabla^2 V = -\rho/\epsilon_0$. $V = \frac{Ae^{-\lambda r}}{r} \Rightarrow \rho = \epsilon_0 A[4\pi\delta^3 - \lambda^2 \frac{e^{-\lambda r}}{r}]$.

3 • Boundary conditions

- $E_{above}^\perp - E_{below}^\perp = \frac{\sigma}{\epsilon_0}$; $E_{\parallel}$ and V continuous. $\frac{\partial V_{out}}{\partial n} - \frac{\partial V_{in}}{\partial n} = -\frac{\sigma}{\epsilon_0}$. Pillbox: flux ($E_a - E_b$) $A = \sigma A/\epsilon_0$. Thin loop: $\oint \vec{E} \cdot d\vec{l} = 0 \Rightarrow E_{\parallel}$ same.
- Conductor: $E = 0$, $\rho = 0$ inside, V const, $\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$, $\vec{E}_{out} = \frac{\sigma}{\epsilon_0} \hat{n}$, pressure $\frac{\sigma^2}{2\epsilon_0}$.
- Words: grounded $\Rightarrow V = 0$. Isolated neutral $\Rightarrow V = \text{unknown const}$, $\oint \sigma = 0$. Held at $V_0 \Rightarrow \text{battery}$, Q free. Far away $\Rightarrow V \rightarrow 0$ or $\rightarrow -E_0 r \cos \theta$. Insulated wall $\Rightarrow \partial V/\partial n = 0$.
- Always also: finite at $r = 0$ / $s = 0$; continuity + jump at every shell.
- Count: each region's unknown constants = number of conditions.

3b • Conductors (the review's 7 facts) + capacitance

- (1) $\vec{E} = 0$ inside; (2) $\rho = 0$ inside; (3) net charge only on the surface; (4) equipotential; (5) $\vec{E} \perp$ surface just outside, $= \frac{\sigma}{\epsilon_0} \hat{n}$; (6) empty cavity: $\vec{E} = 0$ inside it, even in an external field; (7) charge q in a cavity $\Rightarrow -q$ on the cavity wall, $+q$ (+ any net Q) on the outside: the conductor "looks charged".
- $C = \frac{Q}{\Delta V}$ (geometry only). Plates $\frac{\epsilon_0 A}{d}$; spheres $\frac{4\pi\epsilon_0 ab}{b-a}$ (isolated: $4\pi\epsilon_0 R$); coax $\frac{2\pi\epsilon_0 L}{\ln(b/a)}$. Recipe: put $\pm Q$, Gauss for E , $\Delta V = -\int E \cdot d\vec{l}$, divide.

4 • Laplace, uniqueness

- $\nabla^2 V = 0$ where $\rho = 0$: no local max/min; $V(\text{center}) = \text{average over any sphere/circle}$. Earnshaw: no stable equilibrium in empty space.
- **1st theorem**: ρ inside + V on every boundary $\Rightarrow$ unique. **2nd**: conductors with given Q each $\Rightarrow \vec{E}$ unique.
- So any guess (images, separated series) that satisfies Laplace + every BC **is** the answer. Proof: difference $V_3 = V_1 - V_2$ obeys Laplace with 0 on boundary $\Rightarrow$ no max/min $\Rightarrow V_3 = 0$.

5 • Images

- Plane, q at d : $-q$ at $-d$. $\sigma = \frac{-qd}{2\pi(s^2 + d^2)^{3/2}}$, total $-q$, half within $s = \sqrt{3}d$. $F = \frac{kq^2}{(2d)^2}$ (attract).
- Energy: $W = -\frac{kq^2}{4d}$ = **half** the 2-charge value (field only exists on one side). Work to pull off: $+\frac{kq^2}{4d}$.
- Corner 90°: images $-q, -q, +q$. Wedge π/n : $2n-1$ images ($180^\circ/\alpha$ must be an integer). Two parallel planes: infinite row. Line λ : image $-\lambda$.
- **Grounded sphere**: $q' = -\frac{R}{a}q$ at $b = \frac{R^2}{a}$. Derive: $V = 0$ at the near point $a-R$ and far point $a+R \Rightarrow 2$ equations. $F = \frac{kq^2 Ra}{(a^2 - R^2)^2}$.
- $\sigma(\theta) = \frac{-q(a^2 - R^2)}{4\pi R(R^2 + a^2 - 2aR \cos \theta)^{3/2}}$, total $-\frac{qR}{a}$.
- Twists: add charge at the **center**. At V_0 : $+4\pi\epsilon_0 R V_0$. Neutral: $+\frac{qR}{a}$. Net Q : $Q + \frac{qR}{a}$.
- q inside a shell at d : image $-\frac{R}{d}q$ at $\frac{R^2}{d}$ (outside). At V_0 : add V_0 inside.
- Valid only in the region of interest; images never sit in it.

6 • Energy

- $W = \frac{1}{2} \sum q_i V_i$ (V from the others) = $\frac{1}{2} \int \rho V d\tau = \frac{\epsilon_0}{2} \int E^2 d\tau$ (last includes self-energy). Not superposable.
- Ball $\frac{3}{5} \frac{kQ^2}{R}$, shell $\frac{1}{2} \frac{kQ^2}{R}$, pair $\frac{kq_1 q_2}{r}$. $C = \frac{Q}{\Delta V}$, $W = \frac{1}{2} C V^2 = \frac{Q^2}{2C}$; plates $\frac{\epsilon_0 A}{d}$, spheres $\frac{4\pi\epsilon_0 ab}{b-a}$.
- Force: $F = -\frac{dW}{dx}$ at fixed Q . At fixed V the battery does work: $F = +\frac{dW}{dx}$.

7 • Separation: Cartesian

- $V = X(x)Y(y) \Rightarrow \frac{X''}{X} + \frac{Y''}{Y} = 0 \Rightarrow X'' = k^2 X$, $Y'' = -k^2 Y$.
- Direction with **two zero** BCs oscillates: $\sin \frac{n\pi y}{a}$, $k = \frac{n\pi}{a}$. Other direction: $e^{\pm kx}$, $\sinh$, $\cosh$.
- To ∞ : e^{-kx} . Zero at $x = b$: $\sinh k(b-x)$. Zero at 0: $\sinh kx$. Symmetric: $\cosh k(x - \frac{b}{2})$. Insulated walls: $\cos$, plus a $k = 0$ term.
- Fourier: $\int_0^a \sin \frac{n\pi y}{a} \sin \frac{m\pi y}{a} dy = \frac{a}{2} \delta_{nm} \Rightarrow C_n = \frac{2}{a} \int_0^a V_0(y) \sin \frac{n\pi y}{a} dy$. Const: $\frac{4V_0}{n\pi}$ odd n . Linear $V_0 \frac{y}{a}$: $\frac{2V_0(-1)^{n+1}}{n\pi}$.
- Several live faces: one problem per face, add. 3-D: $\sin \sin \sinh(\pi \sqrt{\frac{n^2}{a^2} + \frac{m^2}{b^2}} z)$, const $C_{nm} = \frac{16V_0}{\pi^2 nm}$.
- Series $\rightarrow V_0/2$ at a jump; Gibbs overshoot only in partial sums.

8 • Separation: polar (2-D, no z)

- $\nabla^2 = \frac{1}{s} \partial_s (s \partial_s) + \frac{1}{s^2} \partial_\phi^2$. Try $R = s^\lambda$: $s(sR')' = \lambda^2 R \Rightarrow \lambda = \pm k$.
- $V = a_0 + b_0 \ln s + \sum [s^k (a_k \cos k\phi + b_k \sin k\phi) + s^{-k} (c_k \cos k\phi + d_k \sin k\phi)]$.
- Full circle $\Rightarrow k$ integer, no ϕ term. Origin in region: drop s^{-k} , $\ln s$. ∞ : drop s^k ($\ln s = \text{net } \lambda$).
- Wedge walls 0 at $\theta = 0, \alpha$: $\sin \frac{n\pi\theta}{\alpha}$, $k = \frac{n\pi}{\alpha}$. Zero arc at a : $(\frac{r}{a})^k - (\frac{a}{r})^k$. Fourier: $\frac{2}{\alpha} \int_0^\alpha$.
- One wall V_0 , other 0, no arcs: $V_0 \theta/\alpha$. Tip $|E| \propto r^{\pi/\alpha-1}$ (dies if $\alpha < \pi$, blows up if $\alpha > \pi$).
- Pipe in $E_0 \hat{x}$: $-E_0(s - \frac{R^2}{s}) \cos \phi$, $\sigma = 2\epsilon_0 E_0 \cos \phi$.

9 • Separation: spherical / Legendre

- $V = \sum (A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}}) P_\ell(\cos \theta)$. Try r^n : $n(n+1) = \ell(\ell+1) \Rightarrow n = \ell$ or $-(\ell+1)$. **Not** $r^{-\ell}$.
- Inside: A only. Outside: B only ($+E_0 r \cos \theta$). Shell region $a < r < b$: both.
- $P_0 = 1$, $P_1 = x$, $P_2 = \frac{3x^2-1}{2}$, $P_3 = \frac{5x^3-3x}{2}$, $P_4 = \frac{35x^4-30x^2+3}{8}$; $P_\ell(1) = 1$, $P_\ell(-1) = (-1)^\ell$, $P_2(0) = -\frac{1}{2}$.
- $x^2 = \frac{1}{3} + \frac{2}{3} P_2$, $x^3 = \frac{3}{5} P_1 + \frac{2}{5} P_3$, $\cos 2\theta = -\frac{1}{3} + \frac{4}{3} P_2$, $\sin^2 \theta = \frac{2}{3} - \frac{2}{3} P_2$.
- $c_\ell = \frac{2\ell+1}{2} \int_{-1}^1 f(x) P_\ell dx$ ($x = \cos \theta$). Shell at $V_0(\theta) = \sum c_\ell P_\ell$: in $c_\ell(\frac{r}{R})^\ell$, out $c_\ell(\frac{R}{r})^{\ell+1}$; $\sigma = \frac{\epsilon_0}{R} \sum (2\ell+1) c_\ell P_\ell$; $Q = 4\pi\epsilon_0 R c_0$.
- $\sigma = \sum \sigma_\ell P_\ell$ given: $A_\ell = \frac{\sigma_\ell}{(2\ell+1)\epsilon_0 R^{\ell-1}}$, $B_\ell = A_\ell R^{2\ell+1}$.
- Axis trick: $V(z)$ on axis $\rightarrow$ expand in z^ℓ or $z^{-(\ell+1)} \rightarrow$ replace with $r^\ell P_\ell$, $r^{-(\ell+1)} P_\ell$.
- Total Q only from $\ell = 0$; $\vec{E}(0)$ only from $\ell = 1$; far field from lowest nonzero ℓ .

Recipes (setup is most of the points)

- **Which method:** symmetry $\Rightarrow$ Gauss; charge + flat/round metal $\Rightarrow$ images; box/slot $\Rightarrow$ Cartesian; long wedge/pipe $\Rightarrow$ polar; sphere with $V(\theta)$ or $\sigma(\theta) \Rightarrow$ Legendre; $\vec{E}$ given $\Rightarrow$ divergence + origin check.
- **Separation:** (1) region; (2) numbered BCs; (3) ansatz + sign of constant; (4) zero BCs $\Rightarrow$ basis, quantized k ; (5) superpose; (6) Fourier/Legendre on the live boundary; (7) check a limit / symmetry / corner.
- **Images:** region; BCs on the metal; place images outside the region; verify every BC $\Rightarrow$ uniqueness; $\sigma = -\epsilon_0 \partial_n V$; force = images on q ; energy = half.
- **Write partial results even if stuck:** BC list + general solution earn most of the credit.

Ex 1 • Given $\vec{E}$, find ρ and V

$$\vec{E} = \frac{A}{r^2} \hat{r} \quad (r < R), \quad \frac{AR^2}{r^4} \hat{r} \quad (r > R).$$

- $r < R$: $r^2 E_r = A \text{ const} \Rightarrow \rho = 0$ for $r \neq 0$, but $4\pi\epsilon_0 r^2 E_r \rightarrow 4\pi\epsilon_0 A \Rightarrow \rho \supset 4\pi\epsilon_0 A \delta^3(\vec{r})$.
- $r > R$: $\rho = \frac{\epsilon_0}{r^2} \frac{d}{dr} \frac{AR^2}{r^2} = -\frac{2\epsilon_0 AR^2}{r^5}$. At R : E continuous $\Rightarrow \sigma = 0$.
- Check: cloud = $\int_R^\infty \rho 4\pi r^2 dr = -4\pi\epsilon_0 A \Rightarrow$ total 0, matches $r^2 E \rightarrow 0$.
- $V_{out} = \int_r^\infty E = \frac{AR^2}{3r^3}$; $V_{in} = V(R) + \int_r^R E = \frac{A}{r} - \frac{2A}{3R}$.

Ex 2 • Superposed Gauss

Infinite cylinder radius a (+ ρ , along z) through a ball radius $R > a$ (- ρ).

- Pieces: cyl $\frac{\rho}{2\epsilon_0}(x, y, 0)$ inside; ball $-\frac{\rho}{3\epsilon_0}(x, y, z)$ inside.
- Overlap: $\frac{\rho}{\epsilon_0}(\frac{x}{6}, \frac{y}{6}, -\frac{z}{3})$; check $\nabla \cdot \vec{E} = 0 \checkmark$.
- $z = 0, a < s < R$: $E_s = \frac{\rho a^2}{2\epsilon_0 s} - \frac{\rho s}{3\epsilon_0}$, zero at $s = \sqrt{3/2}a$.
- Axis $z > R$: cylinder gives $0 \Rightarrow E_z = -\frac{\rho R^3}{3\epsilon_0 z^2}$.

Ex 3 • Charge over a grounded plane

- BCs: $V = 0$ on $z = 0$; $V \rightarrow 0$ far. Image $-q$ at $z = -d$.
- $V = kq[\frac{1}{\sqrt{s^2+(z-d)^2}} - \frac{1}{\sqrt{s^2+(z+d)^2}}]$, $\sigma = -\epsilon_0 \partial_z V|_0 = \frac{-qd}{2\pi(s^2+d^2)^{3/2}}$.
- $\int \sigma da = -q$. $F = \frac{kq^2}{4d^2}$ down. $W = \int_\infty^d F dz = -\frac{kq^2}{4d}$ (half of $-\frac{kq^2}{2d}$).

Ex 4 • Sphere held at V_0 + charge q at a

- BCs: (1) $V = V_0$ on $r = R$; (2) $V \rightarrow 0$ at ∞ .
- Images: $q' = -\frac{aR}{a}$ at $\frac{R^2}{a}$, plus $q_0 = 4\pi\epsilon_0 R V_0$ at the center (V_0 everywhere on the sphere).
- $Q_{sphere} = 4\pi\epsilon_0 R V_0 - \frac{qR}{a}$; σ grounded $\sigma + \frac{\epsilon_0 V_0}{R}$.
- $F = \frac{qRV_0}{a^2} - \frac{kq^2 Ra}{(a^2 - R^2)^2}$ (+ = away). $F = 0$ at $a = 2R \Rightarrow V_0 = \frac{8}{9} \frac{kq}{R}$, unstable.

Ex 5 • Cartesian slot

Plates $y = 0, a$ grounded; end $x = 0$ at V_0 ; open to $x \rightarrow \infty$.

- BCs: (1) $V(x, 0) = 0$ (2) $V(x, a) = 0$ (3) $V \rightarrow 0, x \rightarrow \infty$ (4) $V(0, y) = V_0$.
- (1)(2) $\Rightarrow \sin \frac{n\pi y}{a}$; (3) $\Rightarrow e^{-n\pi x/a}$; (4) Fourier $\Rightarrow C_n = \frac{4V_0}{n\pi}$ odd.
- $V = \frac{4V_0}{\pi} \sum_{\text{odd } n} \frac{1}{n} e^{-n\pi x/a} \sin \frac{n\pi y}{a} = \frac{2V_0}{\pi} \tan^{-1} \frac{\sinh(\pi y/a)}{\sinh(\pi x/a)}$.
- Closed at $x = b$ (grounded): replace e^{-kx} by $\frac{\sinh k(b-x)}{\sinh kb}$.

Ex 6 • Polar wedge (Spring '26 P1)

$a < r < b, 0 < \theta < \alpha$; walls + inner arc grounded, $r = b$ at V_0 .

- BCs: $R(a) = 0, \Theta(0) = 0, \Theta(\alpha) = 0, \Phi(b, \theta) = V_0$.
- $\Theta'' = -k^2 \Theta \Rightarrow \sin k\theta, k = \frac{n\pi}{\alpha}; R = C r^k + D r^{-k}, R(a) = 0 \Rightarrow (\frac{r}{a})^k - (\frac{a}{r})^k$.
- $\Phi = \sum C_n \sin \frac{n\pi\theta}{\alpha} [(\frac{r}{a})^k - (\frac{a}{r})^k], C_n = \frac{4V_0}{n\pi[(b/a)^k - (a/b)^k]}$ odd n .
- Polar = Cartesian with $u = \ln r$: the bracket is $2 \sinh(k \ln \frac{r}{a})$.

Ex 7 • Shell at $V_0 \cos^2 \theta$ (Legendre)

- $\cos^2 \theta = \frac{1}{3} P_0 + \frac{2}{3} P_2$.
- $V_{in} = V_0[\frac{1}{3} + \frac{2}{3} \frac{r^2}{R^2} P_2], V_{out} = V_0[\frac{R}{3r} + \frac{2}{3} \frac{R^3}{r^3} P_2]$.
- $\sigma = \frac{\epsilon_0 V_0}{R} [\frac{1}{3} + \frac{10}{3} P_2], Q = \frac{4}{3} \pi \epsilon_0 R V_0, \vec{E}(0) = 0$ (no $\ell = 1$).

Ex 8 • Metal sphere in $E_0 \hat{z}$

- BCs: $V(R) = 0; V \rightarrow -E_0 r \cos \theta$ far.
- Far field is pure $\ell = 1 \Rightarrow V = -E_0 r \cos \theta + \frac{B_1 \cos \theta}{r^2}; V(R) = 0 \Rightarrow B_1 = E_0 R^3$.
- $\sigma = -\epsilon_0 \partial_r V|_R = 3\epsilon_0 E_0 \cos \theta$. Held at V_0 : add $\frac{RV_0}{r}$. Pipe version: R^2/s , factor 2.

Ex 9 • Shell with $\sigma = k \cos \theta$

- BCs: finite at 0; $\rightarrow 0$ at ∞ ; V continuous; $\partial_r V_{out} - \partial_r V_{in} = -\frac{k \cos \theta}{\epsilon_0}$.
- $V_{in} = A r \cos \theta, V_{out} = \frac{AR^3 \cos \theta}{r^2}$; jump: $-2A - A = -\frac{k}{\epsilon_0} \Rightarrow A = \frac{k}{3\epsilon_0}$.
- Inside: uniform $\vec{E} = -\frac{k}{3\epsilon_0} \hat{z}$.

Ex 10 • Energy of a uniform ball, two ways

- $\frac{\epsilon_0}{2} \int E^2: \frac{\epsilon_0}{2} [\int_0^R (\frac{kQr}{R^3})^2 + \int_R^\infty (\frac{kQ}{r^2})^2] 4\pi r^2 dr = \frac{3}{5} \frac{kQ^2}{R}$.
- $\frac{1}{2} \int \rho V$ with $V_{in} = \frac{kQ}{2R} (3R^2 - r^2)$: same.

Ex 11 • Neutral isolated sphere + q at a

- BCs: $V = \text{const}$ (unknown) on the sphere; $\oint \sigma = 0; V \rightarrow 0$ far.
- Images: $-\frac{qR}{a}$ at $\frac{R^2}{a}$ (makes $V = 0$) and $+\frac{qR}{a}$ at the center (restores $Q = 0$, keeps V const).
- $F = \frac{kq^2 R}{a} [\frac{a^2}{(a^2 - R^2)^2} - \frac{1}{a^2}]$ toward the sphere; sphere's $V = \frac{kq}{a}$.

Ex 12 • Charge in a 90° grounded corner

- q at (x_0, y_0) ; walls $x = 0, y = 0$. Images: $-q$ at $(-x_0, y_0), -q$ at $(x_0, -y_0), +q$ at $(-x_0, -y_0)$.
- $F_x = kq^2 [-\frac{1}{4x_0^2} + \frac{x_0}{4(x_0^2 + y_0^2)^{3/2}}]$ (same with $x \leftrightarrow y$). Valid only in $x, y > 0$.

Ex 13 • Charge inside a grounded shell

- q at $d < R$: image $-\frac{qR}{d}$ at $\frac{R^2}{d}$ (outside). $F = \frac{kq^2 R d}{(R^2 - d^2)^2}$ toward the nearest wall.
- Inner surface carries $-q$ total. Shell at V_0 : same inside plus V_0 (same force).

Ex 14 • Box with two live faces

- Square pipe $0 \leq x, y \leq a$: $y = 0$ and $y = a$ at V_0 , sides grounded.
- Sides zero $\Rightarrow \sin \frac{n\pi x}{a}$. Symmetric in $y \Rightarrow \cosh \frac{n\pi(y-a/2)}{a}$. $C_n \cosh \frac{n\pi}{2} = \frac{4V_0}{n\pi}$ (odd).
- Center: by symmetry $V_0/2$ (this + the rotated problem = all faces at V_0).

Classic results to quote

- Nested metal shells: inner surface $-q_{enc}$, outer $Q_{shell} + q_{enc}$. A cavity's field depends only on its own charge.
- Line λ at d over a grounded plane: $V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{s_-}{s_+}$.
- $\frac{1}{|\vec{r}-\vec{r}'|} = \sum_{\ell} \frac{r_\ell'}{r_\ell^{\ell+1}} P_\ell(\cos \gamma)$ (on the axis this is the axis trick).

Traps + integrals

- $d\tau = r^2 \sin \theta dr d\theta d\phi$; σ/ϵ_0 (metal) vs $\sigma/2\epsilon_0$ (sheet); $r^{-(\ell+1)}$; images = half energy; $\sinh$ in the non-zero direction; check units and limits.
- $\int \frac{dx}{(x^2+a^2)^{3/2}} = \frac{x}{a^2 \sqrt{x^2+a^2}}, \int \frac{x dx}{(x^2+a^2)^{3/2}} = \frac{-1}{\sqrt{x^2+a^2}}, \int \frac{dx}{\sqrt{x^2+a^2}} = \ln(x + \sqrt{x^2+a^2}), \int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$.
- $\int_0^\pi \sin^3 = \frac{4}{3}, \int_0^\pi \sin \theta \cos^2 \theta = \frac{2}{3}, \int_0^a \sin^2 \frac{n\pi y}{a} = \frac{a}{2}, \int_0^a \sin \frac{n\pi y}{a} dy = \frac{a(1-(-1)^n)}{n\pi}$.